End-to-End Setup: Microsoft Sentinel & Automated Response

Objective:

✅ Set up Microsoft Sentinel for threat detection
✅ Configure workbooks to visualize attack patterns
✅ Use Logic Apps to send alerts to Teams
✅ Automatically block IPs in response to brute force attacks

PART 1: Setting Up Microsoft Sentinel

STEP 1: Create an Azure Log Analytics Workspace

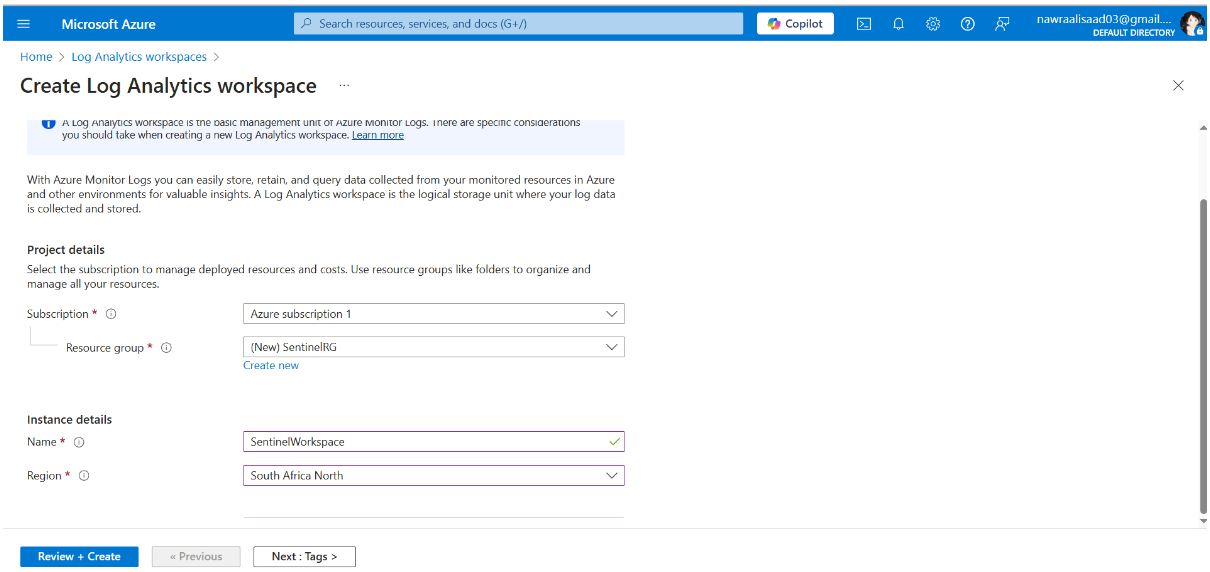
Microsoft Sentinel **needs a Log Analytics Workspace** to store logs.

1️⃣ **Go to** [Azure Portal]()
2️⃣ **Search for** `Log Analytics Workspaces` in the top bar
3️⃣ **Click** “Create”
4️⃣ **Fill in details**:

- **Subscription**: Choose your Azure subscription
- **Resource Group**: Click “Create new” and name it `SentinelRG`
- **Workspace Name**: Set it as `SentinelWorkspace`

- **Region**: Choose the closest data center

5️⃣ **Click** “Review + Create”



STEP 2: Enable Microsoft Sentinel

Now, we’ll **add Microsoft Sentinel** to your Log Analytics Workspace.

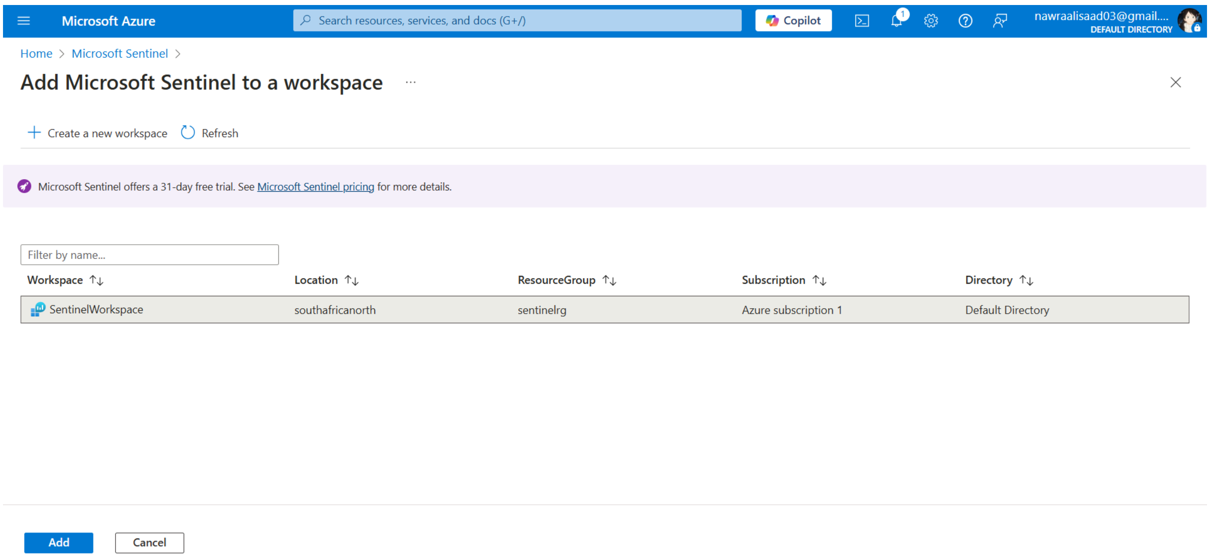
1️⃣ **Go to** [Azure Portal]

2️⃣ **Search for** `Microsoft Sentinel`

3️⃣ **Click** “Add”

4️⃣ **Select** your `SentinelWorkspace`

5️⃣ **Click** “Add Microsoft Sentinel”



✅ **Microsoft Sentinel is now active!**

PART 2: Setting Up a Virtual Machine to Simulate an Attack

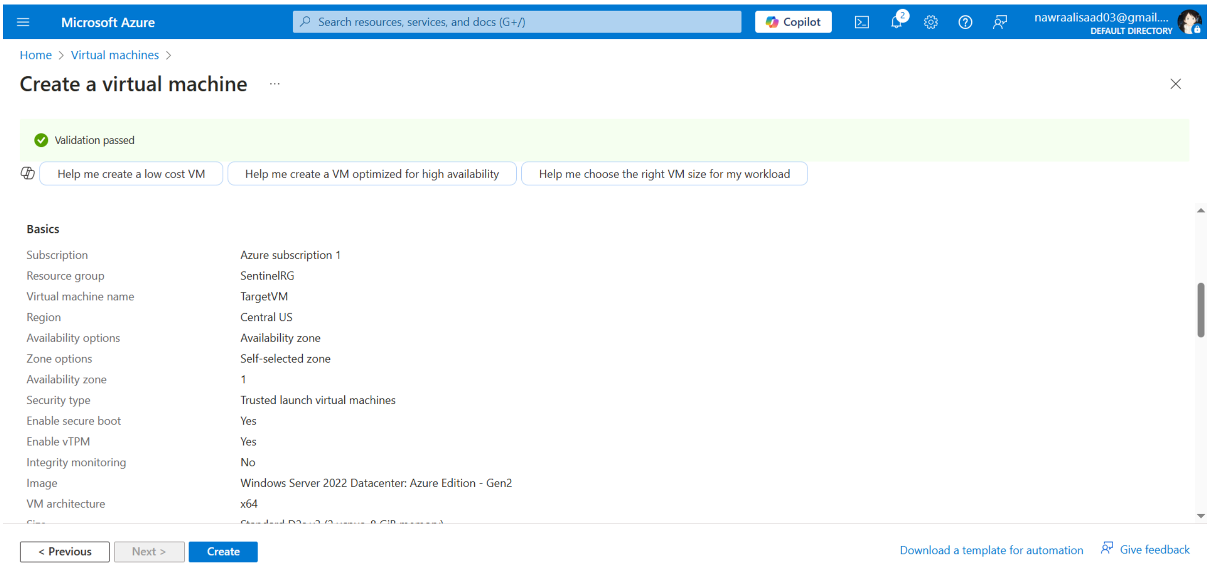
STEP 3: Create a Windows Virtual Machine

We need a **target machine** for the Brute Force Attack.

1️⃣**Go to** Azure Portal > **Virtual Machines**
2️⃣**Click** “Create” > “Azure Virtual Machine”
3️⃣**Configure VM Details**:

- **Resource Group**: Use `SentinelRG`
- **VM Name**: `TargetVM`
- **Region**: Choose the same region as Sentinel
- **Image**: Select `Windows Server 2022`
- **Size**: Choose `B2ms`
- **Username & Password**: Set up credentials
  4️⃣**Click** “Next: Disks” > Keep defaults
  5️⃣**Click** “Next: Networking”
- **Public IP**: Yes (we need it for RDP attacks)

- **Inbound Rules**: Allow **RDP (3389)**
  6️⃣ **Click** “Review + Create”



✅ **Your Virtual Machine is now running!**

STEP 4: Simulate a Brute Force Attack on the VM

Now, we **generate failed login attempts** to trigger Sentinel alerts.

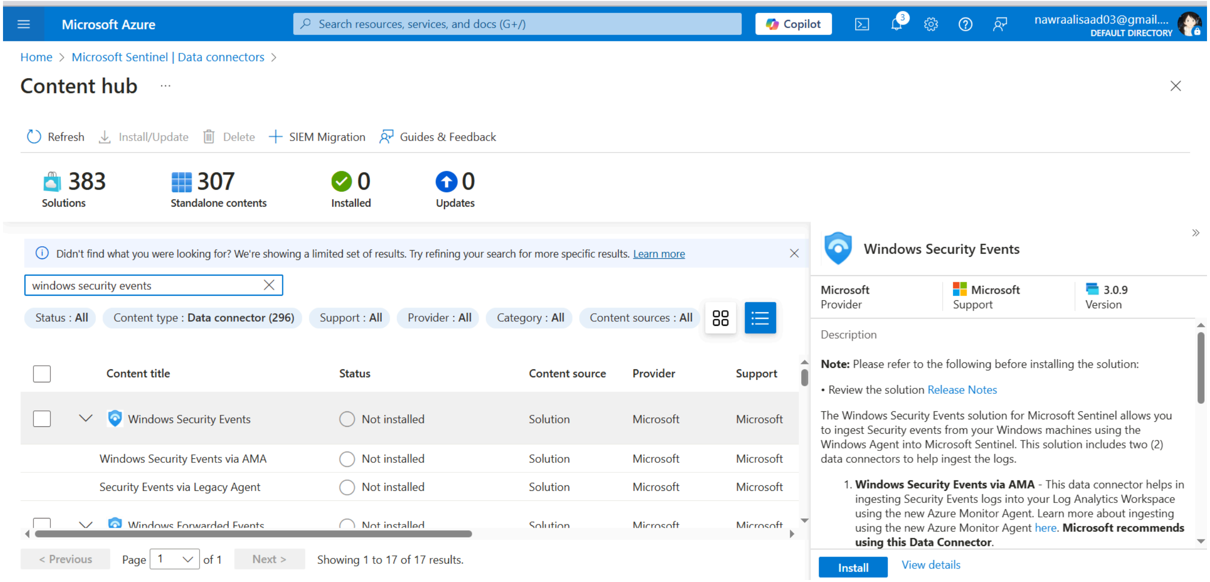
1️⃣ **Copy the Public IP Address** of `TargetVM`
2️⃣ **Open** `Remote Desktop Connection (RDP)` on your computer
3️⃣ **Enter the IP Address** and click “Connect”
4️⃣ **Try logging in with the wrong password multiple times**
5️⃣ **Repeat this at least 10 times in 5 minutes**

🚨 **This generates Event ID 4625 (Failed Login Attempts)**

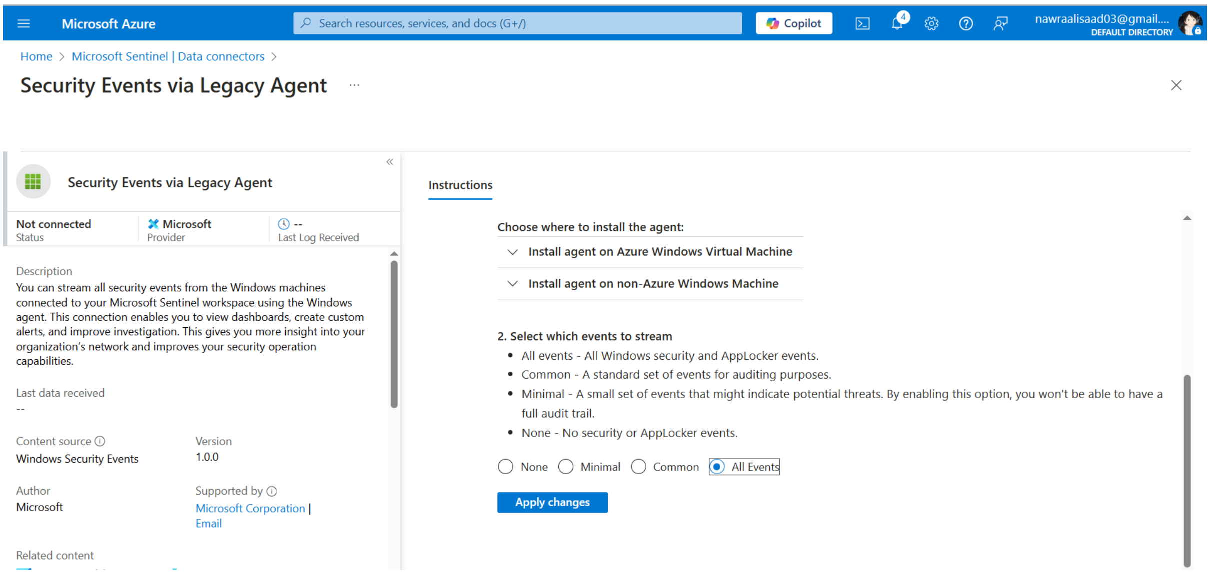
PART 3: Connecting Microsoft Sentinel to Data Sources

STEP 5: Connect Windows Security Logs to Sentinel

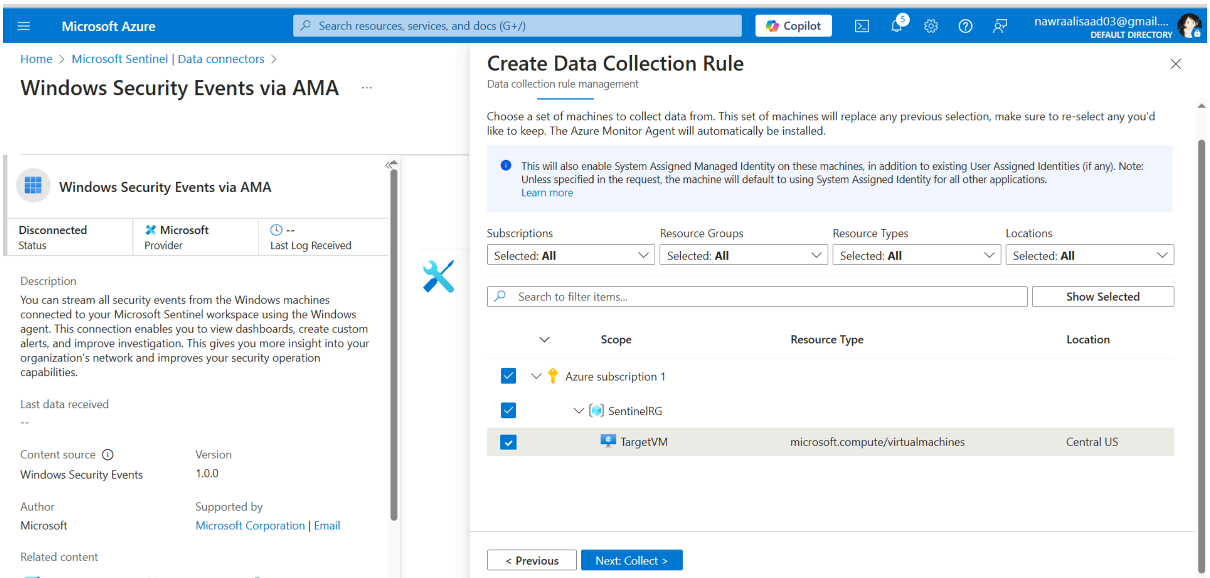
1️⃣ **Go to** `Microsoft Sentinel`
2️⃣ **Click** `Data Connectors`
3️⃣ **Search for** `Windows Security Events`



4 Click `Open Connector Page`



5 Follow the steps to connect your `TargetVM` logs to Sentinel

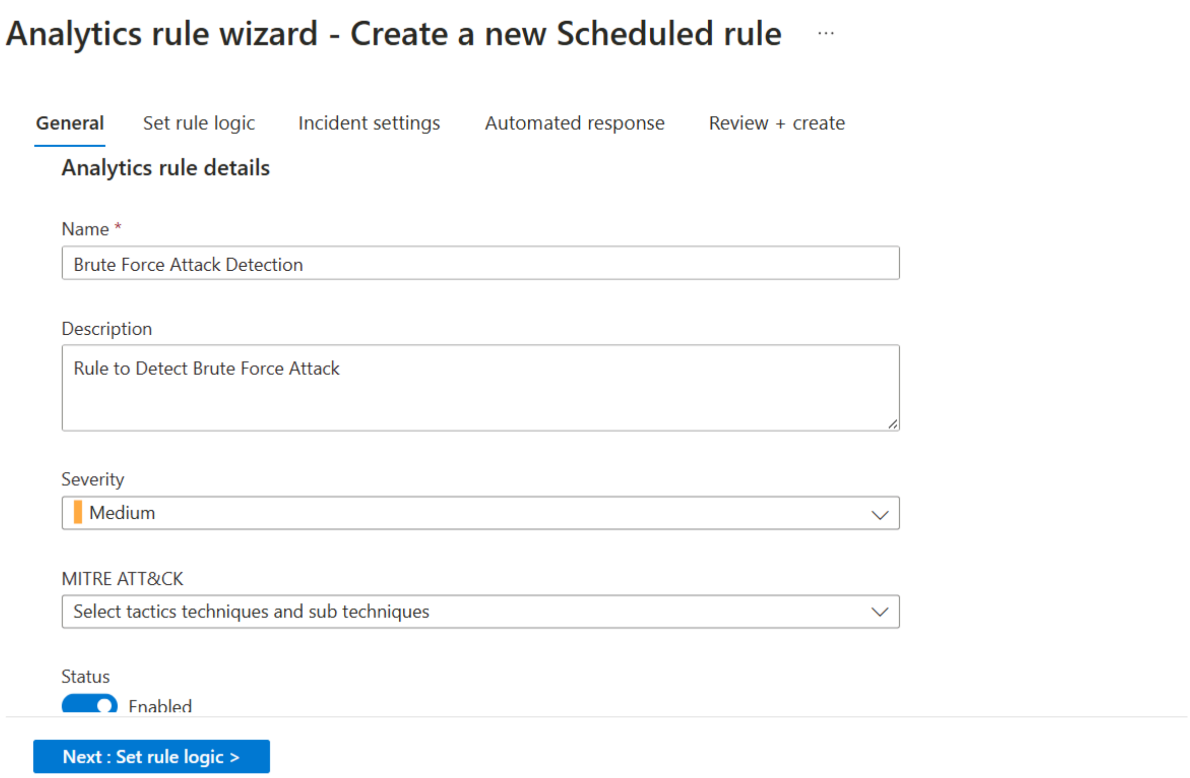


✅ **Sentinel is now collecting security events!**

PART 4: Creating a Detection Rule for Brute Force Attacks

STEP 6: Create an Analytics Rule to Detect Brute Force Attacks

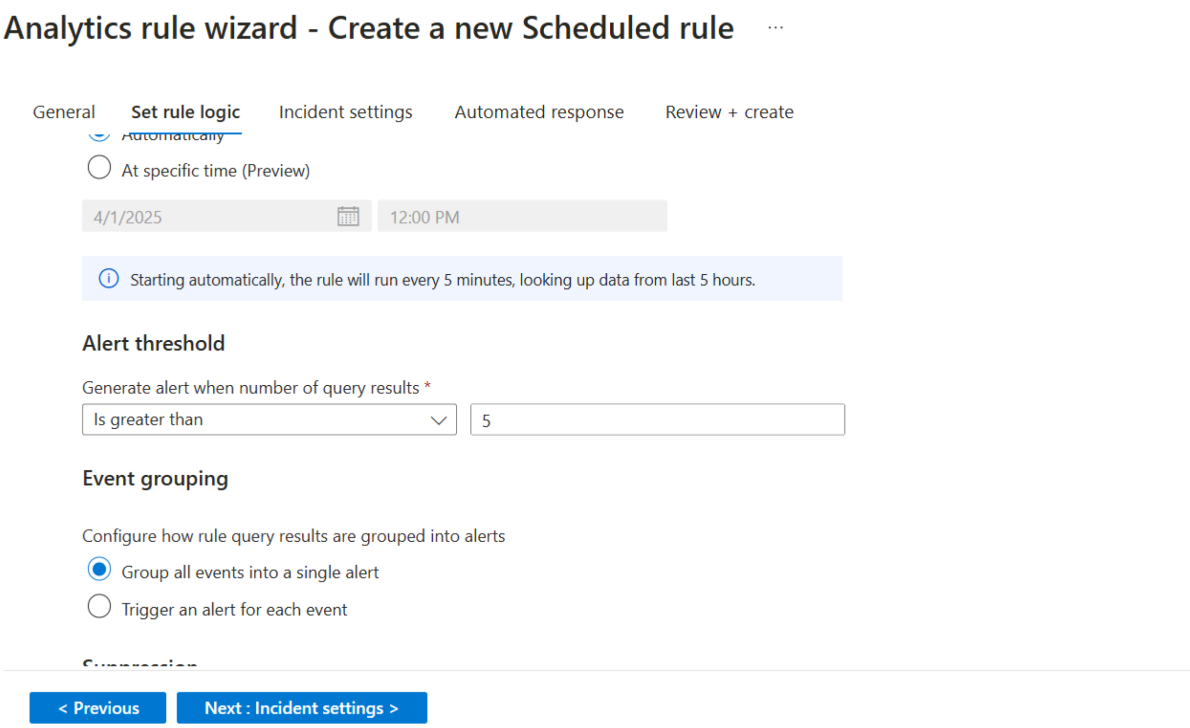
1️⃣ **Go to** `Microsoft Sentinel` > `Analytics`
2️⃣ **Click** `Create` > `Scheduled Query Rule`
3️⃣ **Enter Rule Name**: `Brute Force Attack Detection`



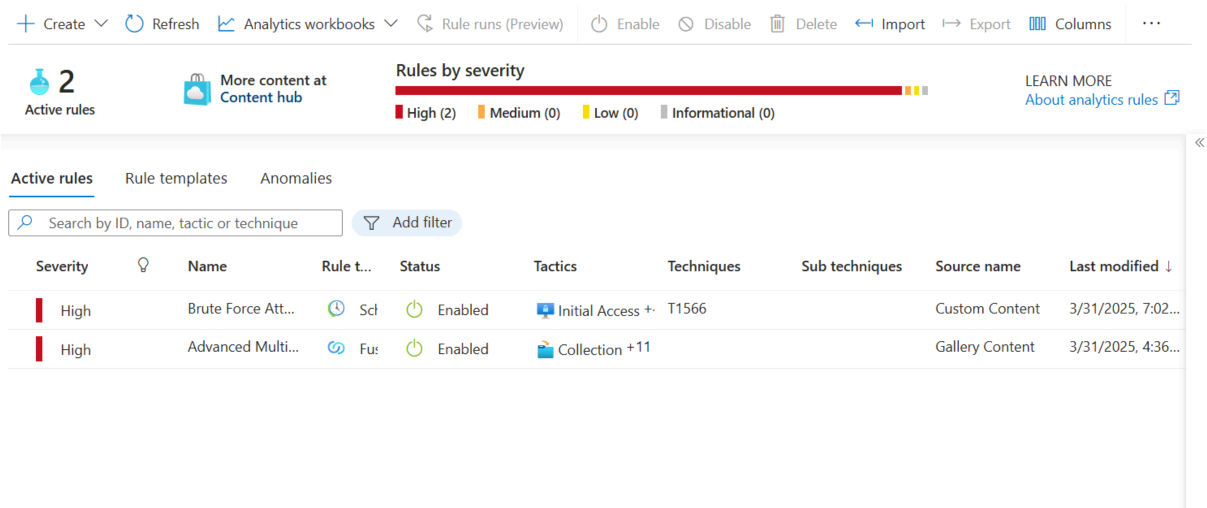
4️⃣ **Use this KQL Query**:

```
SecurityEvent
| where EventID == 4625
| project TimeGenerated, EventID, WorkstationName, Computer,
Account, LogonTypeName, IpAddress
| extend AccountEntity = Account
| extend IpEntity = IpAddress
```

5️⃣ **Set Trigger**: Alert if `FailedLogins > 5`



6️⃣ **Enable the Rule**



✅ **Now, Sentinel will alert you when a Brute Force Attack happens!**

Step 4: Configure Workbooks for Visualization

1. In **Microsoft Sentinel**, go to **Workbooks**

2. Click **Create New**
3. Add a **Visualization Tile**

Choose **Query** and paste:

```
SecurityEvent
| where EventID == 4625
| summarize FailedAttempts = count() by IPAddress,
bin(TimeGenerated, 5m)
| order by FailedAttempts desc
```

4. Click **Apply Changes**
5. Save as **"Brute Force Dashboard"**

📌 **You now have a live dashboard showing attack trends.**

PART 5: Automating Threat Response & ChatGPT Report

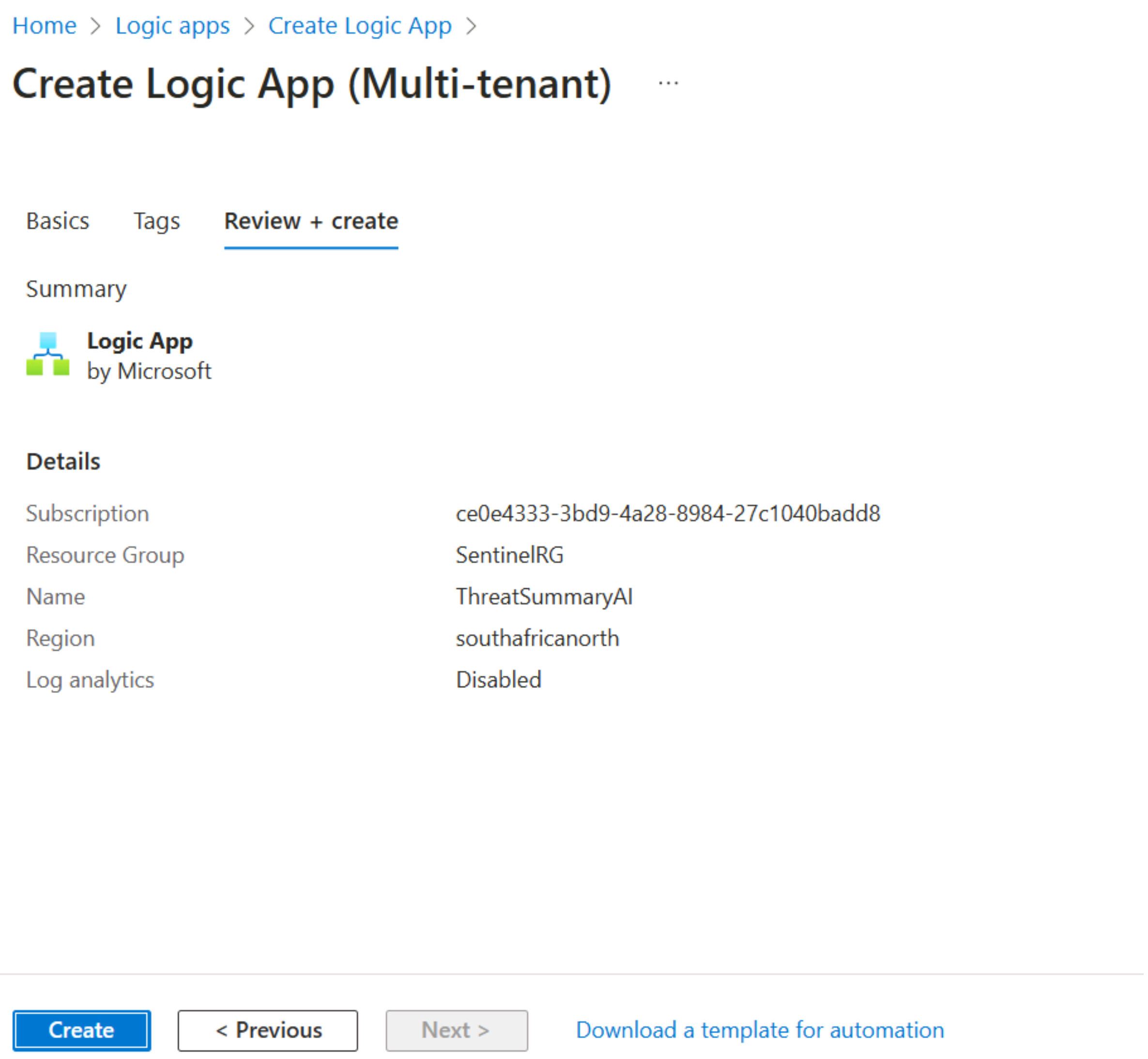
STEP 7: Create a Logic App to Integrate Sentinel with ChatGPT

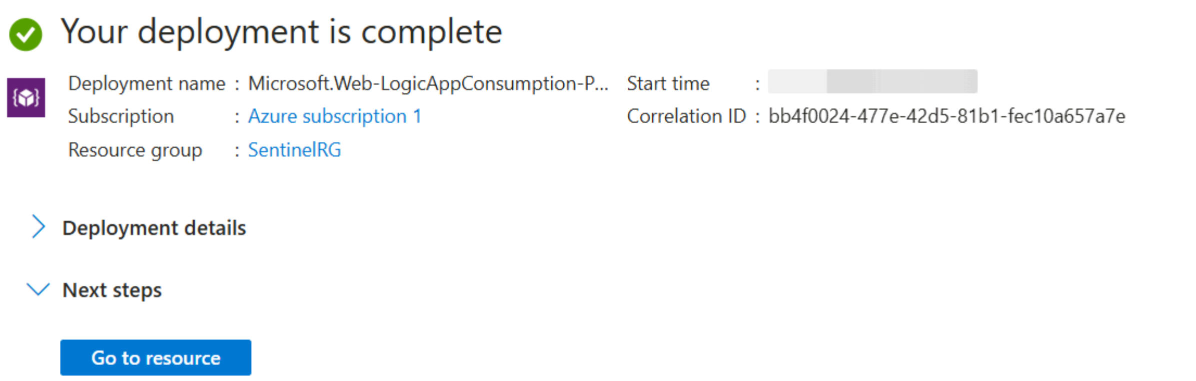
Why? This will automatically summarize the attack details and send them to the security team.

🔹 Create the Logic App

1️⃣**Go to** `Azure Portal`
2️⃣**Search for** `Logic Apps`
3️⃣**Click** "Create"
4️⃣**Set Name**: `ThreatSummaryAI`
5️⃣**Resource Group**: Use `SentinelRG`
6️⃣**Region**: Choose the same region as Sentinel

7️⃣ **Click** `Review + Create`





✅ **Now, let's build the workflow!**

STEP 8: Configure Logic App to Send Alerts to ChatGPT

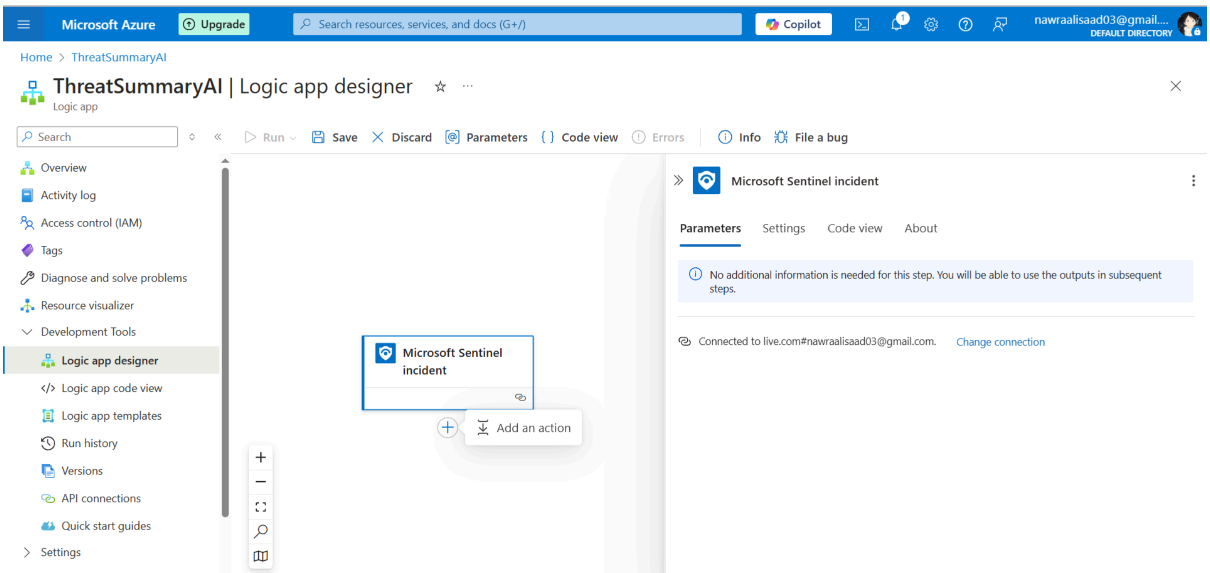
🔹 Add Sentinel as a Trigger

1️⃣ Open `ThreatSummaryAI` Logic App
2️⃣ Click `Logic App Designer

PART 6: Configuring Logic App to Send Alerts to ChatGPT

STEP 9: Add Microsoft Sentinel as a Trigger

1️⃣ **Open** `ThreatSummaryAI` Logic App
2️⃣ **Click** `Logic App Designer`
3️⃣ **Click** `Blank Logic App`
4️⃣ **Click** `+ New Step`
5️⃣ **Search for** `Microsoft Sentinel`
6️⃣ **Select "When an incident is created"**



✅ **This will trigger the Logic App whenever a security incident occurs in Sentinel!**

Step 7: Extract Incident Details

1. Click **"New Step"**

2. Search for **Microsoft Sentinel**
3. Select **"Get Incident"**
4. In **Incident ID**, select `Incident ID` from **Dynamic content**
5. Click **Save**

📌 **This fetches details like the attacker's IP, severity, and timestamp.**

STEP 10: Send Incident Details to ChatGPT

Now, we need to **send the incident data to ChatGPT** for a summary.

1️⃣**Click** `+ New Step`
2️⃣**Search for** `HTTP`
3️⃣**Select "HTTP" action**
4️⃣**Configure the HTTP request**:

- **Method**: `POST`
- **URL**: `https://api.openai.com/v1/chat/completions`

Headers:

json
CopyEdit
```
{
  "Authorization": "Bearer YOUR_OPENAI_API_KEY",
  "Content-Type": "application/json"
}
```

-

Body:

json
CopyEdit
```
{
  "model": "gpt-4",
  "messages": [
    {
      "role": "system",
```

```
      "content": "You are a cybersecurity analyst. Generate a
summary report of the detected security incident."
    },
    {
      "role": "user",
      "content": "Incident Details: {IncidentDescription}"
    }
  ]
}
```

-

5️⃣ **Click** `Save`

✅ **Now, every time Sentinel detects a threat, ChatGPT will summarize it!**

STEP 11: Send ChatGPT's Summary via Email or Teams

Option 1: Send via Email

1️⃣ **Click** `+ New Step`
2️⃣ **Search for** `Office 365 Outlook`
3️⃣ **Select "Send an email (V2)"**
4️⃣ **Fill out the details**:

- **To:** Security Team Email
- **Subject:** `[URGENT] Automated Threat Report`
- **Body:** Use the output from ChatGPT

Option 2: Send via Microsoft Teams

1️⃣ **Click** `+ New Step`
2️⃣ **Search for** `Microsoft Teams`
3️⃣ **Select "Post a message (V3)"**
4️⃣ **Choose the security channel**
5️⃣ **Use ChatGPT's response as the message**

✅ **Now, Sentinel incidents will be summarized by ChatGPT and sent via Email or Teams!**

FINAL TEST: Simulate an Attack & Get a Report

Now, let's **test everything**:

1️⃣**Try the Brute Force Attack on your VM again**
2️⃣**Check Sentinel** → You should see a **new incident**
3️⃣**Check the Logic App** → It should **trigger automatically**
4️⃣**Check your Email/Teams** → You should receive a **ChatGPT-generated summary**